

TikhonovFenichelReductions.jl: A systematic approach to geometric singular perturbation theory

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Summary

Natural systems often evolve on different characteristic time scales, i.e. the rates of processes differ substantially in orders of magnitude (Hek, 2010). Singular perturbation theory is a mathematical toolbox that translates this feature to ODE models of such systems and allows to perform dimensionality reduction when only one characteristic time scale (typically the slow one describing the long term behaviour) is of interest (Wechselberger, 2020). This is reflected by the presence of a small parameter $\varepsilon > 0$, which can be interpreted as the conversion rate between the two time scales. Roughly speaking, the dynamics in slow time $\tau = \varepsilon t$ can be approximated by substituting the fast part of the system with its steady state, i.e. for the slow processes the fast changes appear instant. More precisely, we obtain the *reduced system* (or *reduction* in short) as the limit

$$\begin{aligned} \dot{u} &= g(u, v) \\ \varepsilon \dot{v} &= h(u, v) \end{aligned} \xrightarrow{\varepsilon \rightarrow 0} \begin{aligned} \dot{u} &= g(u, v), & u &\in \mathbb{R}^s \\ 0 &= h(u, v), & v &\in \mathbb{R}^r \end{aligned} \quad (1)$$

with $\dot{} = d/d\tau$ as stated in Tikhonov's theorem (1952; Theorem 1.1 in Verhulst, 2007), which is defined on the so-called slow manifold $M_0 = \{(u, v) \mid h(u, v) = 0\}$. For autonomous systems the results by Fenichel (1979) allow a clear geometric interpretation — hence the name geometric singular perturbation theory (GSPT) — and guarantee that for $\varepsilon > 0$ sufficiently small the reduction captures the behaviour of the full system.

In practice GSPT can mitigate the realism-complexity trade-off, as modellers can work with the reduction instead of the full system, which is likely more feasible. Because the reduction is still embedded in the full system as a special case defined by the time scales, one retains the original interpretability. This allows to follow the modelling paradigm: Start with a complex but realistic model, reduce later.

The algebraic approach recently developed by Goeke & Walcher (and colleagues) allows to work with GSPT systematically and is coordinate-free, i.e. we do not rely on the standard form (1) with a time scale separation of components. Instead, we consider systems of the form

$$x' = f^{(0)}(x) + \varepsilon f^{(1)}(x) + \mathcal{O}(\varepsilon^2), \quad x \in \mathbb{R}^n, \quad (2)$$

where $f^{(0)}$ contains the fast processes and $f^{(1)}$ the slow ones. Note that system (1) can also be stated in fast time t as

$$\begin{aligned} u' &= \varepsilon g(u, v) \\ v' &= h(u, v) \end{aligned} \quad (3)$$

where $' = d/dt$. Thus, we can translate (2) to (3) by setting $x = (u, v)^T$, $f^{(0)} = (0, h)^T$ and $f^{(1)} = (g, 0)^T$, but the reverse requires finding a suitable coordinate transformation, which is not always feasible. However, it turns out that this is not needed explicitly to obtain the reduced system, which can be computed using (4).

35 The main benefit of this framework is however that it allows to find all critical parameters
36 admitting a reduction for polynomial or rational ODE systems of the form (2). Thus, modellers
37 do not have to rely on prior knowledge to find suitable time scale separations as they can
38 systematically consider all of them. This can be achieved by evaluating necessary conditions for
39 the existence of a reduction (Goeke, 2013; Goeke et al., 2015; Goeke & Walcher, 2013, 2014).
40 TikhonovFenichelReductions.jl is a Julia (Bezanson et al., 2017) package implementing
41 this approach for polynomial ODE systems. Apelt & Liebscher (2025) provide a showcasing
42 example and more detailed explanations.

43 Statement of need

44 The ad-hoc approach to singular perturbation theory requires prior knowledge about
45 a suitable time scale separation and substantial mathematical effort to compute the
46 reduction. The algebraic approach due to Goeke & Walcher systematically yields *all*
47 possible reductions for a given polynomial ODE system, which enables modellers to consider
48 reductions for several scenarios defined by the time scale separations (Goeke et al., 2015).
49 TikhonovFenichelReductions.jl makes the required computations easily accessible (even for
50 non-expert users).

51 The author is not aware of any publicly available implementation of this theory, but there exist
52 an implementation for a related approach for computing invariant manifolds by Roberts (n.d.,
53 1997).

54 Software design

55 TikhonovFenichelReductions.jl is implemented in Julia due to its flexibility, the use of
56 multiple dispatch and the availability of the feature-rich computer algebra system Oscar.jl
57 (Decker et al., 2025; The OSCAR Team, 2025).

58 Core features are the search for critical parameters admitting a reduction, so-called *Tikhonov-*
59 *Fenichel Parameter Values (TFPVs)*, and the computation of the corresponding reduced
60 systems. Crucially, this requires various computations with multiple symbolic representations
61 (i.e. different polynomial rings, rational function fields and matrix spaces) and therefore parsing
62 of data between the corresponding types in Oscar.jl, which TikhonovFenichelReductions.jl
63 performs hidden away from the user. Thus, the user mostly works in an object-oriented manner
64 with the types and methods provided.

65 The package is essentially designed around two types: *ReductionProblem*, which constructs
66 all symbolic data types needed for the search of TFPVs, and *Reduction*, which holds all
67 relevant information for the reduced system and the steps required to compute it. The latter
68 also contains the reduced system and other information parsed to the appropriate types from
69 Oscar.jl, that can be further used, e.g. for a symbolic analysis.

70 Detailed explanations and examples are provided by Apelt & Liebscher (2025) and in the
71 [documentation](#). The core features are summarized in the following.

72 Finding TFPVs

73 The package provides a method to obtain all possible TFPVs implicitly by computing a Gröbner
74 Basis and one to find *slow-fast separations of rates*, i.e. TFPVs with some parameters set to
75 zero. Although the former method is an exhaustive search, the latter is usually better suited in
76 practice as it yields the TFPVs one is typically interested in explicitly, is more efficient, and
77 directly outputs the corresponding slow manifolds (implicitly as affine varieties in phase space).
78 In both cases, one utilizes the correspondence between algebraic and geometric properties of
79 the slow manifold, which renders this an algebraic approach to GSPT.

Computing reductions

Computing a reduction for a slow-fast separation of rates as in Theorem 1 in Goeke & Walcher (2014) requires essentially two steps. First, we need to provide a parametric representation of the slow manifold, which is given as an irreducible component of the affine variety $\mathcal{V}(f^{(0)})$. Then, we need to find a product decomposition $f^{(0)} = P\psi$, where P is a $n \times r$ matrix of rational functions and ψ is a vector of polynomials locally satisfying $\mathcal{V}(\psi) = \mathcal{V}(f^{(0)})$ and $\text{rank } P = \text{rank } D\psi = r$.

With this, the reduced system in the sense of Tikhonov is given by

$$x' = [1_n - P(x)(D\psi(x)P(x))^{-1}D\psi(x)] f^{(1)}(x). \quad (4)$$

For convenience, the packages provides multiple heuristics, which automate the steps required to compute a reduction and allows bulk computation of multiple reductions at once.

Integration with the Julia ecosystem

The input system can be given as a reaction network defined with Catalyst.jl (Loman et al., 2023). Because the reduced systems are represented using types from Oscar.jl, the latter's functions can be used to aid the symbolic analysis. Julia's support for metaprogramming allows to perform further tasks such as a numerical analysis without having to copy or parse code (see e.g. TFRSimulations.jl). For convenience, there are several methods for displaying the output, including printing as L^AT_EX source code via Latexify.jl.

Research impact statement

Time scale separations are widely used in various areas of mathematical modelling (Wechselberger, 2020). However, as far as the author is aware, the systematic approach due to Goeke & Walcher seems to be scarcely adopted, even though it comes with many advantages compared to the ad-hoc approach. This package aims to make the theory more accessible and convenient to use.

In the field of mathematical ecology (which the author is most familiar with), the theory was successfully used by Kruff et al. (2019) and more recently by Apelt & Liebscher (2025). For the model introduced in the former, the application was relatively straightforward, but more complex models as in the latter require some more work. In particular, the method for finding TFPVs that relies on the computation of a Gröbner Basis may fail for complex systems. In this case, TikhonovFenichelReductions.jl may enable and simplify the search for the most common TFPVs. Given the ubiquity of time scale separation techniques in this field alone (see e.g. Abbott et al., 2020; Poggiale & Auger, 2004; Revilla, 2015), the package can be a potentially useful tool for modellers.

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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